DISKO 2

⚠️ **Security Warning**: Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

📋 Challenge Description

Can you find the flag in this disk image? The right one is Linux! One wrong step and it's all gone!

💡 Hints

- How can you extract/isolate a partition?

🧰 Tools Used

- `7z` – Extract compressed files
- `file` – Determine file type
- `strings` – Extract printable character sequences
- `exiftool` – Read metadata
- `fdisk` – View partition tables
- `dd` – Extract specific partitions
- `grep` – Filter output

🔍 Complete Solution

Step 1: Download and Extract

Download the disk image named `disko-2.dd.gz` from the challenge page.

Extract the compressed disk image:

```
7z x disko-2.dd.gz
```

This decompresses the gzipped file and creates the raw disk image `disko-2.dd` .

Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file disko-2.dd
```

Expected output:

```
disko-2.dd: DOS/MBR boot sector; partition 1 : ID=0x83, start-CHS (0x0,32,33), end-CHS
(0x3,80,13), startsector 2048, 51200 sectors; partition 2 : ID=0xb, start-CHS
(0x3,80,14), end-CHS (0x7,100,29), startsector 53248, 65536 sectors
```

The output confirms this is a FAT32 filesystem. This is a common filesystem format used on USB drives and memory cards.

Step 3: Search for Visible Text

Disk images often contain strings embedded directly in the data. Extract all readable character sequences and search for the flag:

```
strings disko-2.dd | grep -i "picoCTF"
```

Output (excerpt):

```
picoCTF{4_P4Rt_1t_i5_d3f931a0}
picoCTF{4_P4Rt_1t_i5_a3930df1}
picoCTF{4_P4Rt_1t_i5_f1d0a339}
picoCTF{4_P4Rt_1t_i5_fad03913}
picoCTF{4_P4Rt_1t_i5_139df3a0}
picoCTF{4_P4Rt_1t_i5_f931d3a0}
picoCTF{4_P4Rt_1t_i5_30da391f}
picoCTF{4_P4Rt_1t_i5_af33091d}
picoCTF{4_P4Rt_1t_i5_9d0331fa}
picoCTF{4_P4Rt_1t_i5_13a03f9d}
```

```
picoCTF{4_P4Rt_1t_i5_3df91a30}
picoCTF{4_P4Rt_1t_i5_39f3ad01}
picoCTF{4_P4Rt_1t_i5_930d1fa3}
picoCTF{4_P4Rt_1t_i5_90da3f31}
picoCTF{4_P4Rt_1t_i5_0ad9133f}
picoCTF{4_P4Rt_1t_i5_3ad039f1}
picoCTF{4_P4Rt_1t_i5_339da10f}
picoCTF{4_P4Rt_1t_i5_d33af901}
picoCTF{4_P4Rt_1t_i5_93d0f3a1}
picoCTF{4_P4Rt_1t_i5_9330afd1}
picoCTF{4_P4Rt_1t_i5_9a3d10f3}
picoCTF{4_P4Rt_1t_i5_d9f033a1}
picoCTF{4_P4Rt_1t_i5_d390f1a3}
picoCTF{4_P4Rt_1t_i5_0ad3139f}
picoCTF{4_P4Rt_1t_i5_fa39d031}
picoCTF{4_P4Rt_1t_i5_90a3f3d1}
ronse paquetes en base xa en requiridos: ${picoCTF{4_P4Rt_1t_i5_903d13af}
ce debcopicoCTF{4_P4Rt_1t_i5_393da1f0}dawanym pytaniom priorytety. Tylko pytania o
pewnym lub wy
Description-gl.UTF-8: Configurar unha rede
empreganpicoCTF{4_P4Rt_1t_i5_1930da3f}escription-gu.UTF-8:
ChoicpicoCTF{4_P4Rt_1t_i5_a0f313d9}k Harf Kilidi), Sa
Extended_picoCTF{4_P4Rt_1t_i5_3d1309af}u d'archivu Debian especific
ExtenpicoCTF{4_P4Rt_1t_i5_30f931da}tse tiedostojen hakemiseen k
 fald kanpicoCTF{4_P4Rt_1t_i5_1a33f09d}. Hvis du ikke kender svaret p
picoCTF{4_P4Rt_1t_i5_913a30df}
...
```

> **Observation**: The `strings` command returns many potential flags. We need to dig deeper.

Step 4: Inspect Metadata

Examine the disk image metadata:

```
exiftool disko-2.dd
```

Expected output:

```
ExifTool Version Number         : 13.50
File Name                       : disko-2.dd
Directory                       : .
```

```
File Size                       : 105 MB
File Modification Date/Time     : 2025:06:13 20:16:28+02:00
File Access Date/Time           : 2026:05:09 13:34:29+02:00
File Inode Change Date/Time     : 2026:05:09 13:34:19+02:00
File Permissions                : -rw-r--r--
Error                           : Unknown file type
```

Nothing suspicious here. However, the `file` command output tells us that the image contains **two partitions**.

Step 5: View Partitions

Use `fdisk` to see the partition structure:

```
fdisk -l disko-2.dd
```

Expected output:

```
Disk disko-2.dd: 100 MiB, 104857600 bytes, 204800 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8ef8eaee

Device      Boot Start    End Sectors Size Id Type
disko-2.dd1       2048  53247   51200  25M 83 Linux
disko-2.dd2      53248 118783   65536  32M  b W95 FAT32
```

This shows multiple partitions, including a **Linux** partition (type `83`).

Step 6: Extract the Linux Partition

We will extract the Linux partition (the one we’re interested in) using `dd`. The parameters are:

- `skip=2048` : number of sectors to skip (start of the partition)
- `count=5120` : number of sectors to copy (size of the partition)

```
dd if=disko-2.dd of=linux_partition.dd bs=512 skip=2048 count=5120 status=progress
```

Output:

```
5120+0 records in
5120+0 records out
2621440 bytes (2.6 MB, 2.5 MiB) copied, 0.0465276 s, 56.3 MB/s
```

Step 7: Analyze the Extracted Partition

Check the type of the extracted partition:

```
file linux_partition.dd
```

Output:

```
linux_partition.dd: Linux rev 1.0 ext4 filesystem data, UUID=9e5121c2-9728-4271-a07c-
d74aabdc093b (extents) (64bit) (large files) (huge files)
```

Examine the metadata:

```
exiftool linux_partition.dd
```

Output:

```
ExifTool Version Number         : 13.50
File Name                       : linux_partition.dd
Directory                       : .
File Size                       : 2.6 MB
File Modification Date/Time     : 2026:05:09 13:59:57+02:00
File Access Date/Time           : 2026:05:09 14:00:13+02:00
File Inode Change Date/Time     : 2026:05:09 13:59:57+02:00
File Permissions                : -rw-r--r--
Error                           : First 1025 bytes of file is binary zeros
```

Now, search for the flag within this partition:

```
strings linux_partition.dd | grep -i picoCTF
```

And boom! We get the flag.

✅ Final Result

The flag is found directly in the disk image:

```
picoCTF{<REDACTED>}
```

> **Note**: The actual flag is redacted here. In the real challenge, the `strings` command will reveal the correct flag.

🧠 Key Concepts

- **Disk Image (DD)**: A raw sector-by-sector copy of a storage device.
- **MBR Partition Table**: A structure that divides a disk into multiple partitions.
- **Partition Extraction with `dd`**: Using `skip` and `count` to isolate a specific partition.
- **EXT4 Filesystem**: Linux native filesystem, commonly used for system partitions.
- **Forensic Analysis**: Inspecting metadata and extracting strings to find hidden data.